TJ Collamore

tjc269@cornell.edu • 978-405-4685 • Portfolio: tjcollamore.com • https://linkedin.com/in/tjcollamore • https://github.com/tjcollamore

EDUCATION

Cornell University — Ithaca, NY
Major: Computer Science, College of Engineering — May 2030
Courses: CS 1110, MATH 1910, ENGRI 1101

Westford Academy — Westford, MA
GPA: 4.27 | **High Honors:** 2022, 2023, 2024, 2025, 2026 — June 2026
Book Award: Clarkson University Leadership & Achievement Book Award 2025
Affiliations: German National Honor Society (2024 - 2026)
AP Credit: Computer Science A (5), Chemistry (5), Statistics (5)

Tufts University Pre-College Engineering with Artificial Intelligence Program — July - August 2025
- Built projects integrating **mechanical systems**, **sensors**, and **machine learning algorithms**, including:
  - Chronic Kidney Disease visualization using **Nearest Neighbors algorithm**
  - Race Car Simulation utilizing **Reinforcement Learning** & **Artificial Neural Networking**

PROJECTS (featured at tjcollamore.com)

Blackjack Game — July 2025 - Present
- Designed and implemented in **Java** with a **JavaFX GUI**, using **Replit** as the development environment
- Focused on independent mastery of **object-oriented design** and **recursion**, and **purposefully limited AI assistance to UI elements** only
- Built **game logic** for an 8-deck shoe, shuffling, hit/stay/double/split/surrender, betting, dealer play, win detection, and bankroll tracking

Race Car Simulation — July - August 2025
- Developed a race car simulation in **Python** where a **neural network** learns optimal steering and acceleration using **reinforcement learning**
- Tuned **reward functions** and **visualized model convergence** throughout the **training process**
- Used **Claude Code** to accelerate development

Tar Spot Image Analysis — September - November 2024
- Wrote a **C# image-processing program** to quantify how fungal tar spot affects chlorophyll concentration in leaves
- Combined biological analysis with **computational modeling** to enhance accuracy of chlorophyll measurement

TRACK / CROSS COUNTRY

Westford Academy Spring and Winter Track & Field, *Captain* — 2022 - 2026
- Main events: 2-mile, 1-mile, 800m
- **Personal Records: 9:51 2-mile, 4:44 1-mile, 2:03 800m**
- 8th-fastest 2-mile in school history
- Led a team of **~70 runners through training & warm-ups**
- **Created & distributed training plans**, recommending training times, splits, and distances for each running group
  - Inspired by those of Stanford University & University of Colorado D1 teams via collaboration with WA alumni
- Commitment: 15 hours/week, 45 miles/week, 28 weeks/year

Westford Academy Cross Country (WA XC), *Captain* — 2022 - 2026
- Competed in all DCL meets, invitationals, and championships
- **Personal Record: 16:47 5k**
- Selected to represent WA XC at the **UMass Lowell Leadership Conference**
  - Met with MA governors, professional runners, & NFL athletes
- Commitment: 15 hours/week, 45 miles/week, 14 weeks/year

INDUSTRY EXPERIENCE & LEADERSHIP

Raytheon, *Job Shadow* — May 2025
- Selected to work with Chief Engineer Michael P. Clark
- Firsthand exposure to advanced defense systems, missile and sensor technologies, and immersive 3D simulation tools

Westford Flag Football Head Coach — July - November 2023
- Led a middle school flag football team to semi-finals and received the **League Sportsmanship Coaching Award**